

Veterinary Drug Handbook

3rd Edition

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Most birds: 250 mg/kg via feed for 5-10 days. Sprinkle on favorite food or add to mash or corn mix. (Clubb 1986)

b) 100 mg/kg IM or IM q8h (Hoeffler 1995)

Reptiles:

For susceptible infections:

- Snakes: 3 - 6 mg/kg IM or SQ daily (McConnell and Hughey 1987)]
- All species: 3 - 6 mg/kg PO, SubQ or IM every 12-24 hours for 2 weeks; not very useful unless used in combination with aminoglycosides. (Gauvin 1993)

Client Information - Unless otherwise instructed by the veterinarian, this drug should be given orally on an empty stomach, at least 1 hour before feeding or 2 hours after. Keep oral suspension in the refrigerator and discard any unused suspension after 14 days. If stored at room temperature, discard unused suspension after 7 days.

Dosage Forms/Preparations/FDA Approval Status/Withholding Times - Veterinary-Approved Products:

Ampicillin Trihydrate Injection Powder for Suspension 10 g & 25 g (of ampicillin) vials.

Polyflex® (Fort Dodge); (Rx) Approved for use in dogs, cats, and cattle. Withdrawal times (cattle; do not treat for more than 7 days): Milk = 48 hours; Slaughter = 6 days.

Ampicillin Sodium for Injection 1 gram & 3 gram vials (of ampicillin)

Amp-Equine® (Pfizer); (Rx) Approved for use in horses not intended for food.

Human-Approved Products:

Ampicillin Sodium Powder for Injection Vials 125 mg, 250 mg, 500 mg, 1 g, 2 g, 10 g; 500 mg, 1 g, and 2 gram piggyback units; *Omnipen-N*® (Wyeth-Ayerst) (Rx); *Polycillin-N*® (Apothecon) (Rx); *Totacillin-N*® (SK-Beecham) (Rx); generic (Rx)

Ampicillin Capsules (as either trihydrate or anhydrous) 250 mg, 500 mg; Many tradenames and generics; (Rx)

Ampicillin (as the trihydrate) Powder for Oral Suspension 25 mg/ml, 50 mg/ml, 100 mg/ml in 20 ml, 80 ml, 100 ml, 150 ml, and 200 ml bottles; Many tradenames; (Rx)

Also available in fixed dose combinations with:

Probenecid: Powder for Oral Suspension: 3.5 g ampicillin (as trihydrate) & 1g probenecid/probenecid (oral); *Polycillin-PRB*® (Apothecon); *Probampacin*®, generic (Rx)

Sulbactam Sodium (injection): 1.5 g (1 g ampicillin sodium/.05 g sulbactam sodium) & 3 g (2 g ampicillin sodium/1 g sulbactam sodium); *Unasyn*® (Roerig) (Rx)

AMPROLIUM HYDROCHLORIDE

Chemistry - A structural analogue of thiamine (vitamin B₁), amprolium hydrochloride occurs as a white or almost white, odorless or nearly odorless powder. One gram is soluble in 2 ml of water and is slightly soluble in alcohol.

Storage/Stability/Compatibility - Unless otherwise instructed by the manufacturer, amprolium products should be stored at room temperature (15-30°C).

Pharmacology - By mimicking its structure, amprolium competitively inhibits thiamine utilization by the parasite. Prolonged high dosages can cause thiamine deficiency in the host and excessive thiamine in the diet can reduce or reverse the anticoccidial activity of the drug.

Amprolium reportedly acts primarily upon the first generation schizont in the cells of the intestinal wall, preventing differentiation of the merozoites. It may also suppress the sexual stages and sporulation of the oocysts.

Uses/Indications - Amprolium has good activity against *Eimeria tenella*, *E. acervulina* in poultry and can be used as a therapeutic agent for these organisms. It only has marginal activity or weak activity against *E. maxima*, *E. mivati*, *E. necatrix*, or *E. brunetti*. It is often used in combination with other agents (e.g., ethopabate) to improve control against those organisms.

In cattle, amprolium has approval for the treatment and prevention of *E. bovis* and *E. zurnii* in cattle and calves. Amprolium has been used in dogs, swine, sheep, and goats for the control of coccidiosis, although there are no approved products in the U.S.A. for these species.

Pharmacokinetics - No information was located for this agent.

Contraindications/Precautions/Reproductive Safety - Not recommended to be used for over 12 days in puppies.

Adverse Effects/Warnings - In dogs, neural disturbances, depression, anorexia, and diarrhea have been reported but are rare and are probably dose-related. See Overdosage section below for treatment recommendations.

AS-1

AS-2

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Overdosage/Acute Toxicity - Amprolium has induced polioencephalomalacia (PEM) in sheep when administered at 880 mg/kg PO for 4-6 weeks and at 1 gram/kg for 3-5 weeks. Erythrocyte production in lambs receiving these high dosages of amprolium also ceased.

A14 It is reported that overdoses of amprolium will produce neurologic symptoms in dogs. Treatment should consist of stopping amprolium therapy and administering parenteral thiamine (1 - 10 mg/day IM or IV).

A11 **Drug Interactions** - Exogenously administered thiamine in high doses may reverse or reduce the efficacy of amprolium.

Doses -

Dogs:

For coccidiosis:

- 100 - 200 mg/kg PO in food or water for 7-10 days. (Morgan 1988), (Kirk 1989), (Greene 1984)
- Prophylaxis: 30 ml of 9.6% solution in one gallon (3.8 L) of drinking water or (not both) 1.25 grams of 20% powder in food to feed 4 pups daily. Give as sole source of food or water for 7 days prior to shipping. Bitches may be given medicated water (as above) as the sole source of water for 10 days prior to whelping. (USPC 1989)
- Prophylaxis: 0.075% solution as drinking water (Matz 1995)

Cattle:

For coccidiosis:

- Treatment: 10 mg/kg PO for 5 days; 5 mg/kg for 21 days for prophylaxis. (Todd, Dipietro, and Guterbock 1986)

Swine:

For coccidiosis:

- Treatment: 25 - 65 mg/kg PO once or twice daily for 3-4 days. (Todd, Dipietro, and Guterbock 1986)
- 100 mg/kg/day in food or water. (Howard 1986)

Sheep & Goats:

For coccidiosis:

- Lambs: 55 mg/kg daily PO for 19 days. (Todd, Dipietro, and Guterbock 1986)

Birds:

- For coccidiosis in pet birds: 2 ml (using the 9.6% solution)/gallon of water for 5 days or longer. Cages should be steam cleaned to prevent reinfection. Supplement diet with B vitamins. Some strains resistant in Toucans and Mynahs. (Clubb 1986)
- For chickens (broilers or layers), turkeys, and pheasants: Refer to individual product instructions.

Monitoring Parameters -

- Clinical efficacy

Dosage Forms/Preparations/FDA Approval Status/Withholding Times -

Veterinary-Approved Products:

Amprolium 1.25% Medicated Crumbles for Top Dressing in 50 lb bags.; *Corid*® 1.25% Crumbles (MSD-AgVet); (OTC) Approved for use in calves. Slaughter withdrawal = 24 hours.

Amprolium 9.6% (96 mg/ml) Oral Solution; *Corid*® 9.6% Oral Solution (MSD-AgVet); (OTC) Approved for use in calves. Slaughter withdrawal = 24 hours

Amprolium 20 % Soluble Powder; *Corid*® 20% Soluble Powder (MSD-AgVet); (OTC) Approved for use in calves. Slaughter withdrawal = 24 hours

There are many combination products (medicated feeds, feed additives) containing amprolium with other therapeutic agents. These products are approved for chickens (broilers only) and/or turkeys. There are some products containing amprolium alone for use in laying hens.

Human-Approved Products: None

AMRINONE LACTATE

Chemistry - Unrelated structurally to cardiac glycosides or catecholamines, amrinone is a bipyridine cardiac inotropic agent. It occurs as a pale yellow, crystalline powder and is insoluble in water and slightly soluble in alcohol. The commercially available injection has a pH adjusted to 3.2-4 and an osmolality of 101 mOsm/L.

Storage/Stability/Compatibility - The commercially available injection should be stored at room temperature and protected from light. It is stable for 2 years after manufacture.

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